KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2024
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

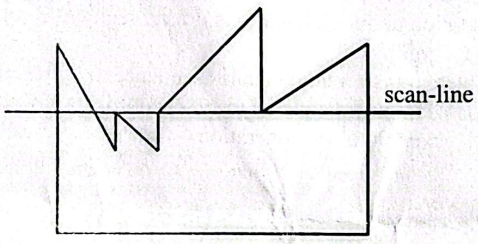
N.B. i) Answer ANY THREE questions from each section in separate scripts.
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iii) The rightmost column indicates course outcomes.

SECTION A

(Answer ANY THREE questions from this section in Script A)

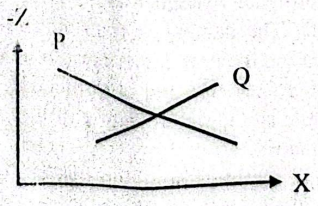
1. a) Suppose, you are designing a CRT monitor for displaying realistic scenes. Choose the appropriate combination of display type (vector/raster) and color production method (beam penetration/shadow mask) for the task and justify your choice. (10) [CO1]
   b) A line segment P_0P_1 is defined by the implicit equation $5x - 8y + 4 = 0$, where the end points are constrained such that $y_{start} = 8$ and $y_{end} = 13$. Using Bresenham's line drawing algorithm, determine the internal points. (12) [CO2]
   c) A graphics system renders two line segments: $L1$ $(-1, 7)$ to $(11, 1)$ and $L2$ $(5, 4)$ to $(9, 12)$. The visible area is restricted to a square shaped clipping window with $(x_{min}, y_{min}) = (1, 1)$ and $(x_{max}, y_{max}) = (9, 9)$. Using Liang-Barsky line clipping algorithm, determine the portion of the line segments that are visible within the window. (13) [CO3]

2. a) Explain how standard ray tracer be modified to generate soft shadow. (08) [CO3]
   b) Apply the Scan-Line Fill algorithm to demonstrate how the given polygon is filled along the specific scan-line. (10) [CO2]

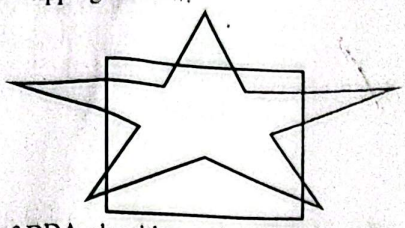


   c) A texture T of size 512×512 is mapped onto a planar surface. You are given a normalized coordinate point on that surface as $P(u, v) = (0.4, 0.6)$. Determine the corresponding texture coordinate $T(x, y)$ to be mapped on P and compute the final color at this point using GL_LINEAR texture filtering method. Given the color of the texels: $T(x, y) = (123, 205, 75)$, $T(x + 1, y) = (200, 150, 100)$, $T(x, y + 1) = (80, 220, 130)$, and $T(x + 1, y + 1) = (160, 90, 240)$. (17) [CO3]

3. a) Explain how the ray tracing method integrates the process of projection, scan conversion and shadow generation. (10) [CO2]
   b) A fractal is generated from an equilateral triangle. In each iteration, every side is replaced using the Koch snowflake construction, and each new edge is transformed into a Cantor edge of the same level. Now, draw the third-generation of the fractal and calculate the dimension of the structure. (12) [CO3]
   c) Given edge-on polygons P and Q in the $(x, -z)$ plane with overlapping (x, y) extents and assuming the view port looks towards the negative z axis. Apply depth sorting algorithm to examine the tests on the polygons and state the final rendering order of them. (13) [CO2]



4. a) Apply Weiler-Atherton polygon clipping algorithm to clip the star shaped polygon using the rectangle as a clipping window. (10) [CO1]



b) State the limitations of DDA algorithm and explain how Bresenham's line drawing algorithm overcomes them? (07) [CO1]

c) Consider a 3D scene where a point light source is located at the center of a sphere of radius 3 with center $C = (-1, 7, -4)$. A light ray emitted from the source intersects the sphere at point p, then strikes a triangle defined by $v_0 = (-5, 1, -5)$, $v_1 = (1, -1, -4)$, $v_2 = (0, 15, -7)$ with approximate surface normal $N = -1.04i + 1.77j + 9.79k$ at point q, and reflects towards the camera at origin $s = (0, 0, 0)$ passing through an image plane at depth 2. Trace the equivalent view ray from camera to the direction $d = (-1, 2, -2)$ and determine the intersection points p and q. (18) [CO3]

SECTION B

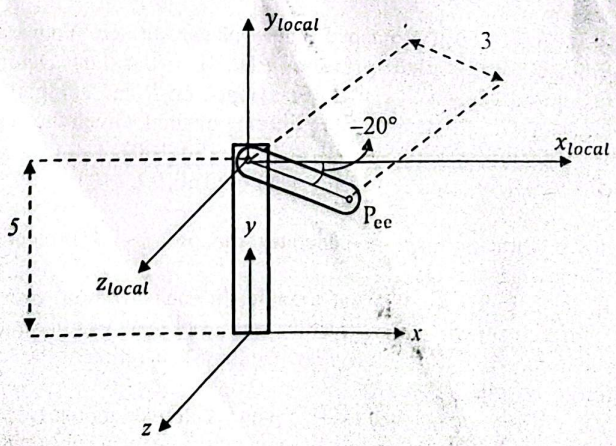
(Answer **ANY THREE** questions from this section in Script B)

5. a) Proof that 3D rotation matrix can be written as $R = \mathbf{I}\cos\theta + \sin\theta\,\mathbf{U} + (1 - \cos\theta)\mathbf{u}\mathbf{u}^T$, where $\mathbf{u}$ is unit axis of rotation, θ = angle of rotation, and $\mathbf{U}$ is skew symmetric matrix satisfying $\mathbf{Uv} = \mathbf{u} \times \mathbf{v}$. Be noted $\mathbf{U}^2 = \mathbf{u}\mathbf{u}^T - \mathbf{I}$. (18) [CO1]

b) Consider the following figure showing a 2 segment robotic arm fixed at origin. Segment 1 has length 5 and placed along y-axis. Segment 2 has length 3, attached to the end of segment 1, and initially it is oriented $-20°$ from its local x-axis. The entire assembly undergoes $+30°$ rotation about global z-axis. Later, segment 2 rotated by $+45°$ about the local y-axis of the joint. (17) [CO3]

Construct the complete transformation matrix sequence to represent the position of the end-effector tip. Also compute the global (x, y, z) coordinates of the tip.

[Hint: The complete operation can be defined as $P_{ee} = T_{global} \cdot T_{L_1} \cdot T_{joint-operation} \cdot T_{init_1} \cdot T_{L_2} \cdot P_{origin}$ where T's are appropriate transformation matrices.]



6. a) A camera in World Coordinate System (WCS) is defined by eye $(7, 8, 9)$, look-at $(1, 1, 1)$, and up $(0, 1, 0)$. The camera is translated (slid) by $(1, -1, 1)$ in its own coordinate system. Determine the new camera origin in WCS after the slide. (13) [CO3]

b) You are given a point $P(2, 4, 5)$, and a projection plane defined by equation $x - 2y + 2z = 3$. Compute isometric projection of the point on the plane. Also, show the isometric projection matrix. (10) [CO1]

c) Derive the general perspective transformation matrix. Verify that it maps (r, t, n) to $(1, 1, -1)$. Symbols have their usual meaning. (12) [CO2]

CamScanner

7. a) State how Ward illumination model estimates anisotropic specular reflection. (07) [CO1]

b) A point $P(1,2,3)$ lies on a surface with normal $\boldsymbol{N} = (0,0,1)$, viewer is at $(1,2,8)$. A monochromatic spot light is positioned at $(4,5,6)$ with light focus vector $\frac{1}{\sqrt{3}}(-1,-1,-1)$ and cutoff angle 30°. Determine whether point P lies inside the spot-light cone. If so, compute the angular attenuation factor with fall off exponent = 2. [Note: Be careful about light vector and focus vector's direction.] (08) [CO2]

c) A triangle in 3D space has vertices and normals: $A(1,1,1), N_A(0,-4,3); B(5,1,1), N_B(0,-3,3); C(3,4,5), N_C(1,-4,2)$. A point $P(3,2,2.33)$ lies inside the triangle. Light and view parameters are: L_{pos} $(6,6,8)$, Eye_{pos} $(6,-2,5)$, $I_L = 1$. You are supposed to render the triangle using Phong shading method, and compute intensity by applying Cook-Torrance illumination model. Compute lerped normal at P using Barycentric interpolation. Then, compute specular intensity at P if surface roughness $m = 0.4$, Fresnel base reflectivity $F_0 = 0.1$. Following equations are given for your convenience, where symbols have their usual meaning. Be sure to clamp negative dot products. (20) [CO3]

$$I_{spec} = I_L \cdot f_r \cdot (N \cdot L), \quad f_r = \frac{DFG}{4(N\cdot L)(N\cdot V)},$$

$$\text{Beckmann Distribution } D = \frac{1}{\pi m^2 (N\cdot H)^4} e,$$

$$\text{Schlick Fresnel } F = F_0 + (1 - F_0)(1 - V \cdot H)^5,$$

$$\text{Cook-Torrance Geometry } G = \min\left(1, \frac{2(N\cdot H)(N\cdot V)}{V\cdot H}, \frac{2(N\cdot H)(N\cdot L)}{V\cdot H}\right).$$

8. a) Neatly write shooting algorithm to compute radiosity of a patch. Why does it converge faster than gathering method? (10) [CO1]

b) A tensor product Bézier surface patch $S(u,v)$ is defined as cubic in u direction and quadratic in v direction. (13) [CO3]

i) Write fully expanded matrix equation to compute $S(u,v)$. Your answer must explicitly show the contents and dimensions of vectors U, V, the blending matrices B_i, B_j and geometry matrix P_{ij}.

ii) Using the components from your matrix expression, write the specific 2D basis function $B_{2,1}(u,v)$ that is used for control point $P_{2,1}$. [1D Bernstein polynomial of degree n at index k is $B_{k,n}(t) = \binom{n}{k} t^k (1-t)^{n-k}$]

c) Consider a quadratic B-spline curve (order $k = 3$) is defined by 4 control points $P_0, \cdots, P_3$ and a non-uniform knot vector $T = \{0, 1, 3, 4, 5, 6, 7\}$. Using the Cox-de Boor recursive formula given below, derive the exact algebraic expression for the basis function $B_{1,3}(u)$. (12) [CO2]

$$B_{k,1}(u) = \begin{cases} 1, & \text{if } u_k \le u \le u_{k+1} \\ 0, & \text{Otherwise} \end{cases}$$

$$B_{k,d}(u) = \frac{u - u_k}{u_{k+d-1} - u_k} B_{k,d-1}(u) + \frac{u_{k+d} - u}{u_{k+d} - u_{k+1}} B_{k+1,d-1}(u), \quad d > 1$$

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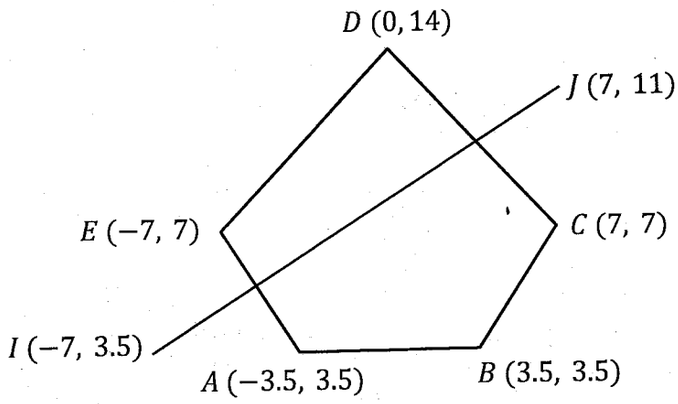
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SECTION A

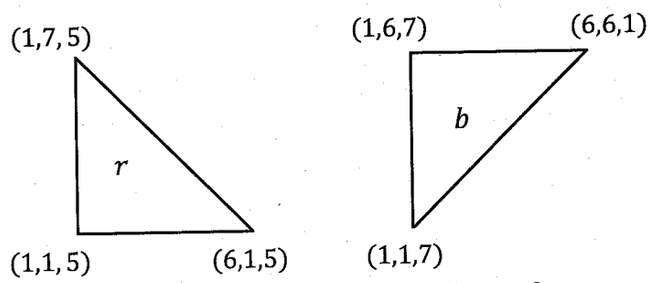
(Answer **ANY THREE** questions from this section in Script A)

1. a) Suppose you are given a task to choose a display system for your project. It needs to be able to draw millions of lines, triangles or characters on the screen. Which display system will you choose? Explain your reasonings. (08) [CO1]
   b) Describe the types of fractals that can be found in the real world. (09) [CO1]
   c) Figure below (not drawn to scale) shows a clipping polygon defined by five vertices. A line IJ is defined by two vertices I and J. Clip the line IJ against the clipping polygon using Cyrus-Beck algorithm and draw the final result after clipping. (18) [CO2]



$$t = \frac{N_i.[P_{E_i} - P_0]}{N_i.D}$$

2. a) What steps are required to scan convert an ellipse using the trigonometric method. (10) [CO2]
   b) The following figure shows two triangles and its vertices. Using Z-buffer, calculate the color of the pixels of color buffer. Show the depth buffer in each step. Assume that the color buffer has 8 (eight) rows and columns. The first triangle is red, denoted by *"r"* and the second one is blue, denoted by *"b"*. Write *"r"* or *"b"* to denote color red and blue, and *"bc"* for background color. (15) [CO2]



   c) Suppose, you are given a task of generating texture coordinates for an actual 3D model of earth. Describe two ways of achieving your goal. (10) [CO1]

3. a) Write an algorithm to determine if a ray intersects a triangle or not. Be sure that it is not the plane containing the triangle but the triangle itself. (12) [CO2]
   b) A line P_0P_1, described by two points $P_0 = (21.2, 12.3)$ and $P_1 = (28.8, 15)$, is given. You have to rasterize the line using Bresenham's line drawing algorithm. Calculate five consecutive points on the line starting from P_0 [excluding P_0]. (15) [CO2]
   c) "On a raster display, diagonal lines feel dimmer than a horizontal or vertical line." Do you agree with the statement? Explain your reasonings. (08) [CO1]

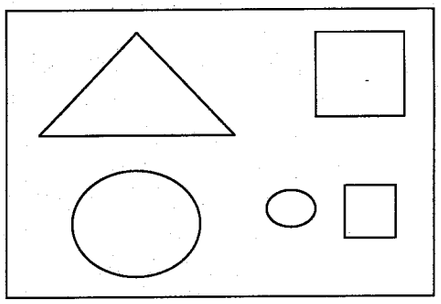
4. a) Describe how mipmapping achieves anti-aliasing. (09) [CO1]

b) Suppose there is a scene consisting of two opaque spheres. A red sphere with center $C_r = (2, 4, -10)$ and radius $R_r = 5$, and a blue sphere with center $C_b = (4, 8, -20)$ and radius $R_b = 3$. We want to calculate the color of a pixel which have direction vector $d = (0.18, 0.36, 0.91)$from the camera origin. Calculate the color of that pixel. For simplicity let's assume there is only white ambient light in the scene. The ray tracing tree shouldn't be deeper than two levels. The background color is black. (20) [CO3]

For your convenience here are some formulas, where the symbols have their usual meanings in this context.

$$t = \frac{-B \pm \sqrt{B^2 - 4C}}{2}; B = 2(s - c).d; C = |s - c|^2 - R^2; r = u - 2(u.n)n$$

c) The following figure shows a 2D slice of a 3D scene. Draw the approximate output of applying (i) KD tree (ii) Octree algorithm to divide the scene into multiple subspaces. (06) [CO2]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Derive the scaling matrix for scaling an object by a scale factor 3 along an arbitrary direction given by vector $u = (1, 2)$ rooted at $(4, 3)$. (08) [CO1]

b) Use quaternion basics to show that (13) [CO2]

$$\boldsymbol{qPq}^{-1} = (s^2 - \boldsymbol{v}^2)\boldsymbol{P} + 2s\boldsymbol{v} \times \boldsymbol{P} + 2(\boldsymbol{v}.\boldsymbol{P})\boldsymbol{v}$$

where **P** is a pure quaternion and $\boldsymbol{q} = [s + \boldsymbol{v}]$ is a unit quaternion.

c) The Camera Coordinate System (CCS) is defined by the basis vectors $\boldsymbol{u}(0.782, 0.569, 0.254)$, $\boldsymbol{v}(0.079, 0.927, -0.367)$ and $\boldsymbol{n}(0.334, 0.668, 0.665)$. Currently the camera is at world origin, and you sequentially pitched $(0^\circ \leq \theta^\circ \leq 90^\circ)$ and rolled $(-90^\circ \leq \beta^\circ \leq 0^\circ)$ the camera. (14) [CO3]

If after transform the basis vectors are $\boldsymbol{u}'(0.425, 0.828, -0.366)$, $\boldsymbol{v}'(-0.0044, 0.166, -0.986)$ and $\boldsymbol{n}'(-0.166, 0.955, -0.246)$, find the pitch, θ and roll, β. [Hint: Following equations are given for your convenience.]

$$a \sin x + b \cos x = c$$
$$x = \sin^{-1}(c/\sqrt{a^2 + b^2}) - \tan^{-1}(b/a) + 2n\pi$$
$$or, x = \pi - \sin^{-1}(c/\sqrt{a^2 + b^2}) - \tan^{-1}(b/a) + 2n\pi$$

6. a) Consider the following perspective matrix defined for projection plane containing point $R_0(x_0, y_0, z_0)$, plane normal $N(n_1, n_2, n_3)$, and projection reference point $P_{prp}(a, b, c)$, with $d = d_0 - d_1 = N.R_0 - N.P_{prp}$. (12) [CO3]

$$\begin{bmatrix} d + an_1 & an_2 & an_3 & -ad_0 \\ bn_1 & d + bn_2 & bn_3 & -bd_0 \\ cn_1 & cn_2 & d + cn_3 & -cd_0 \\ n_1 & n_2 & n_3 & -d_1 \end{bmatrix}$$

Show that, for this perspective transform, vanishing point P_v in the direction vector U can be formulated as $P_v = P_{prp} + \left(\frac{d}{K}\right) U$ where $K = N.U$. Show the derivation steps in details.

b) Explain the importance of using Distribution Term, D and Geometry Term, G in their illumination model by Cook-Torrance. How do they estimate those terms? Show with necessary illustration, reasoning and derivation. (15) [CO1]

c) For the given control points (0,0), (0,1), (4,0), (5,0); use Hermite blending functions to generate a cubic curve $Q(t)$. Find the point on the curve at $t = 0.2$. (08) [CO2]

7. a) Let you are drawing a non-uniform cubic B-spline curve $Q(t)$, with control points $(5,6),(10,25),(13,10),(20,30)$ and $(26,15)$ and knot vector [0,0,0,0,2,4,5,6,7]. Determine the blending functions for the parameter range $2 \leq t \leq 4$. Also find the point on the curve $Q(t)$ at $t = 3.0$. For your convenience, the recursive Cox-deBoor blending equations are given below, where symbols have their usual meanings. (16) [CO1]

$$B_{k,1}(t) = \begin{cases} 1, & if\ t_k \leq t \leq t_{k+1} \\ 0, & Otherwise \end{cases}$$

$$B_{k,d}(t) = C_k B_{k,d-1}(t) + (1 - C_{k+1}) B_{k+1,d-1}(t)$$

$$C_k = \frac{t - t_k}{t_{k+d-1} - t_k}$$

b) Let we want to define a parametric surface patch $S(u,v)$ using Bezier blending functions. Suppose the patch is quadratic and cubic on the parametric directions u and v respectively. How many control points do you need to supply? Represent the patch in tensor notation, clearly stating each item. Particularly show the Bezier basis matrices for different directions by generating from Bernstein polynomial equation. (11) [CO3]

c) Let a plane P contains the origin and a unit vector **u** is orthogonal to the plane. A reflection across P is a linear transform that maps each point **x** to its mirror image directly across P. Prove that this reflection transform for such a plane can be represented by a 3×3 matrix $\boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^T$. Write out the matrix in component form too. (08) [CO2]
[Hint: Use idea similar to the rotation matrix derivation used by Rodrigues, where $v_{ref} = v_{\perp} - v_{\parallel}$]

8. a) Use Gauss-Seidal iterative approach of solving radiosity equation to find the radiosities of each of the patches having following information. Approximate values up to four decimal places. (20) [CO2]

Patch	1	2	3	4
Emissive Power	0	0	0.6	0.8
Area	0.4	0.6	0.5	0.2
Reflectivity	0.8	0.6	0.3	0.3

		To			
		1	2	3	4
From	1	–	0.5	0.6	0.3
	2	–	–	0.4	0.1
	3	–	–	–	0.2

Form factors

b) Explain why Gouraud surface rendering method produces much better color approximation than constant shading method, especially for curved surfaces represented by polygonal mesh. (07) [CO3]

c) Derive the equations for computing delta form factor of a pixel located at back face of a hemicube. (08) [CO1]

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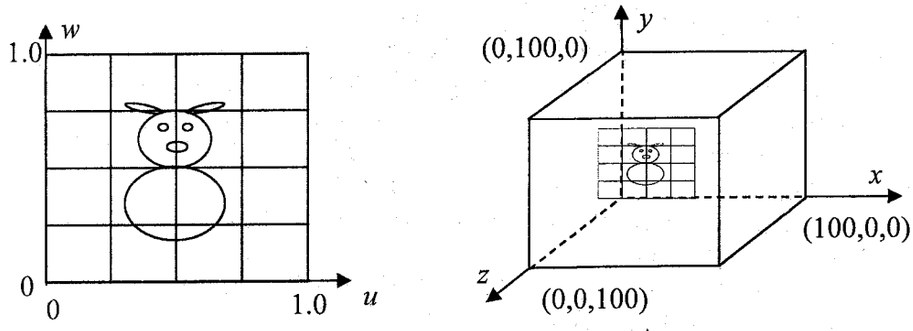
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SECTION A

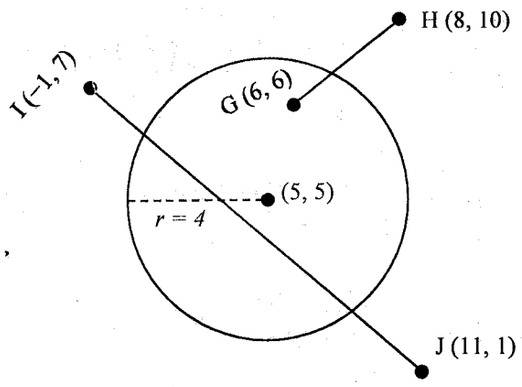
(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07) [CO1]

b) Consider the following figure and find the linear functions that map the normalized image onto a square area of 50 × 50 in the middle of the front face of the cubic object. (12) [CO1]

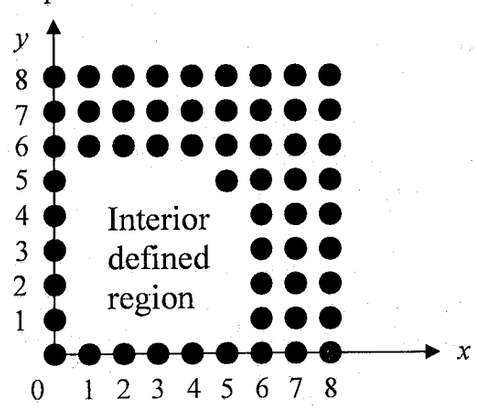


c) Use the Liang-Barsky algorithm to clip the lines in following figure. The clipping window is an axis aligned square that tightly encompasses the given circle. (16) [CO2]



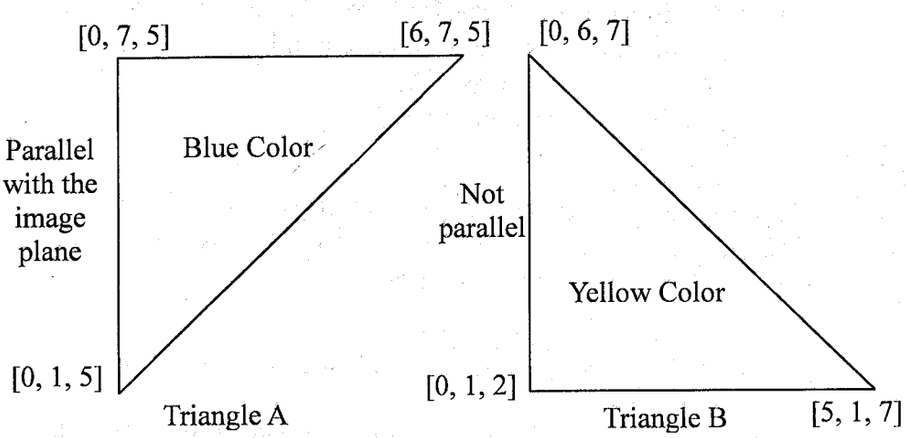
2. a) Indicate which raster locations would be chosen by Bresenham's algorithm when scan converting a line from point (1, 9) to (6, 2). (12) [CO1]

b) How would a flood-fill algorithm fill the region shown in following figure, using the 8-connected definition for region pixels? (13) [CO3]



c) When eight-way symmetry is used to obtain a full circle from pixel coordinates generated for the 0° to 45° or the 90° to 45° octant, certain pixels are set or plotted twice. This phenomenon is sometimes referred to as overstrike. Identify where overstrike occurs. (10) [CO1]

3. a) Using a 8×8 display, show how the z-buffer algorithm would determine the color of each pixel for the given triangles A (Blue Color) and B (Yellow Color) in following figure. Write 'B' for blue color and 'Y' for yellow color in the color buffer. (15) [CO2]



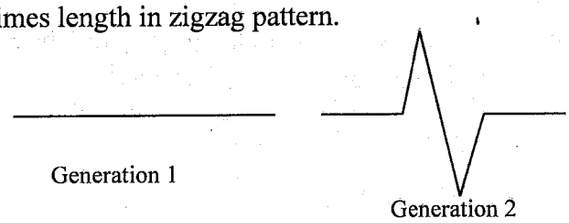
b) Explain Sutherland-Hodgeman polygon clipping algorithm. Mention it's limitations. (12) [CO3]

c) Briefly discuss how the roughness of a surface can be simulated using texture mapping. (08) [CO1]

4. a) Let, $Q(t) = (1 + t^2)p_1 + (5t - 5t^2)p_2 + t^2p_3$, and $R(t) = (1 + t^2)^2p_3 + (5t - 5t^2)p_4 + t^2p_5$ be the parametric equations of two curves, where t is the parameter and P_1, P_2, P_3, P_4, P_5 are the points on the curves. Determine whether $Q(t)$ and $R(t)$ are C^0, C^1, and G^2 continuous. (07) [CO1]

b) A Bezier curve is to be drawn given the control points $P_1(0,0)$, $P_2(3,3)$, $P_3(8,1)$, $P_4(10,2)$. Calculate the coordinates of the points on the curve corresponding to the parameter $t = 0.2, 0.4, 0.5, 0.6, 0.8$. Draw a rough sketch of the curve and show coordinates of the various points on it. (16) [CO3]

c) Two generation of a fractal is shown below, where middle one third of original curve is replaced with 4-times length in zigzag pattern. (12) [CO3]



i) Draw the third generation,
ii) Determine length of n-th generation of the fractal,
iii) Determine the dimension of the fractal.

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Find the composite transformation matrix (affine matrix) that transforms the triangle with vertices $A(1,2)$, $B(5,2)$, and $C(3,5)$ into vertices $A'(-1,-2)$, $B'(-4,-5)$, $C'(2,-2)$ respectively. (14) [CO3]

b) Determine the reflection of a 2D point $P(5,9)$ about a line $L(t) = (2,11) + (3,9)t$. Let you are given with the following reflection transformation matrix: (12) [CO2]

$$M_L = \frac{1}{m^2+1}\begin{bmatrix} 1-m^2 & 2m & -2bm \\ 2m & m^2-1 & 2b \\ 0 & 0 & m^2+1 \end{bmatrix}$$

c) Show that the path of a ray transmitted through a surface is $\mathbf{t} = \mathbf{z} - (\sqrt{1 - |\mathbf{z}|^2})\mathbf{n}$, where $\mathbf{z} = \frac{\eta_1}{\eta_2}(\mathbf{u} - (\mathbf{u}.\mathbf{n}))\mathbf{n}$, $\mathbf{u}$ is an incoming ray, $\mathbf{n}$ is the surface normal, and η_1, η_2 are the refractive indices of the media. (09) [CO3]

6. a) The following matrix is a general parallel projection matrix with direction of projection (DOP) onto an arbitrary plane with plane normal N, passing through P_0. Now, determine the parallel projection of the point $(2,3,4)$ if DOP is $(1,1,1)$ and projection plane is $2x + y + z = 8$. Symbols have their usual meanings in the given matrix. (12) [CO2]

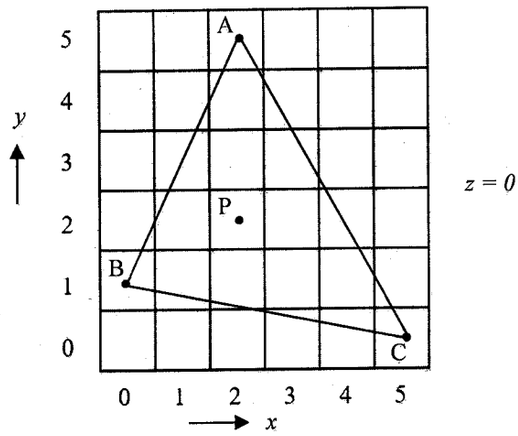
$$Par_{V,N,P_0} = \begin{bmatrix} d_1 - an_1 & -an_2 & -an_3 & ad_0 \\ -bn_1 & d_1 - bn_2 & -bn_3 & bd_0 \\ -cn_1 & -cn_2 & d_1 - cn_3 & cd_0 \\ 0 & 0 & 0 & d_1 \end{bmatrix}$$

b) Suppose you want to align vector $\bar{p} = (0, 1, 1)$ with the vector $\bar{q} = (1, 0, 1)$ by rotating around the axis $(-1, -1, 1)$. Consider the given quaternion-based rotation matrix $M_R(\theta)$, where symbols have their usual meanings. What will be the angle of rotation? (12) [CO3]

$$M_R(\theta) = \begin{bmatrix} 1 - 2b^2 - 2c^2 & 2ab - 2sc & 2ac + 2sb \\ 2ab + 2sc & 1 - 2a^2 - 2c^2 & 2bc - 2sa \\ 2ac - 2sb & 2bc + 2sa & 1 - 2a^2 - 2b^2 \end{bmatrix}$$

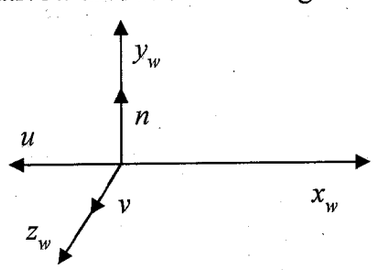
c) Derive a transformation matrix to convert a frustum shape eye-space view volume in a canonical view volume in normalized device coordinate space. (11) [CO1]

7. a) Consider a triangular fragment ABC as shown below. Let, the vertex normals are $N_A = (0, 0, 1)$, $N_B = (0, 0.2, 1)$, and $N_C = (0, -0.2, 1)$. If the light source is at $(7, 3, 4)$, viewer is at $(-6, 3, 3)$ with light color $(0.9, 0.9, 1)$, surface materials reflectivity $(0.5, 0.6, 0.4)$ and you are rendering the surface with Phong shading then determine the diffuse color at point P. (14) [CO1]



b) How can you generate soft shadows using ray tracing? Explain. (06) [CO2]

c) The current position of a camera is at $(10, 20, -15)$, looking at $(35, 40, 20)$, and head up is along $(3i + 4j - 5k)$. Find the transformation matrix that would align the camera looking direction along with positive y-axis, and the direction that points to the right of the camera along with the negative x-axis as shown in figure below. (15) [CO1]



8. a) Let a ray is defined by $\mathbf{s} = (0, 0, 0)$, $\mathbf{d} = (-1/\sqrt{14}, 3/\sqrt{14}, -2/\sqrt{14})$. (15) [CO2]

i) Determine if the ray intersects the triangle defined by the points $(-5, 1, -5)$, $(1, -1, -4)$, and $(0, 18, -7)$,

ii) Also find the transmitted ray considering the triangle as transparent surface with $\eta = 1.5$.

b) Consider following two tables showing information regarding four patches and their form factors. Determine radiosity of each patch. (15) [CO1]

Patch No	1	2	3	4
Reflectivity, ρ_i	0.2	0.5	0.6	0.4
Emissive intensity, E_i	0.8	0	0	0
Patch area	0.2	0.6	0.5	0.7

	1	2	3	4
1	1	0.6	0.6	0.3
2	–	1	0.3	0.1
3	–	–	1	0.1
4	–	–	–	1

Form Factor, F_{ij}

Approximate values up to three decimal places.

c) Given that a hemicube is defined at origin with the z axis vertical. What is the delta form factor of the hemicube pixel on the right face $x = 1$, if $(y_{min}, y_{max}, z_{min}, z_{max}) = (0.55, 0.65, 0.35, 0.45)$. (05) [CO3]

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B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

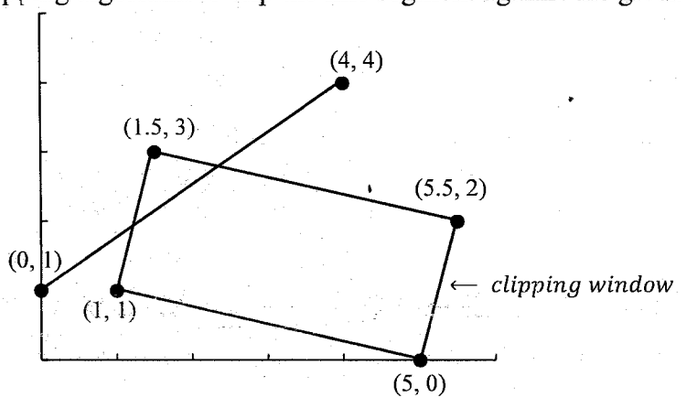
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly explain the different functionalities of 3D computer graphics pipeline. (10)
   b) Show that two successive reflections about any two lines in the xy plane that go through the origin is equivalent to a rotation about the origin. (12)
   c) Use quaternion mathematics to derive a 3D rotation matrix about a unit vector $\boldsymbol{u}$. (13)

2. a) Derive a perspective projection matrix if the center of projection is at (a, b, c), the projection plane normal $N = n_1 i + n_2 j + n_3 k$ and reference point $R_0(x_0, y_0, z_0)$. (13)
   b) Determine the oblique projection of a VCS point $P(2, 2, 5)$ onto a view plane with $N = k$ and $R_0(1, 0, 1)$. Use azimuth = 30° and elevation = 45°. (10)
   c) Consider the following figure showing a rectangular clipping window and a line segment. Apply Cyrus-Beck line clipping algorithm to clip the line segment against the given window. Find intersection points. (12)

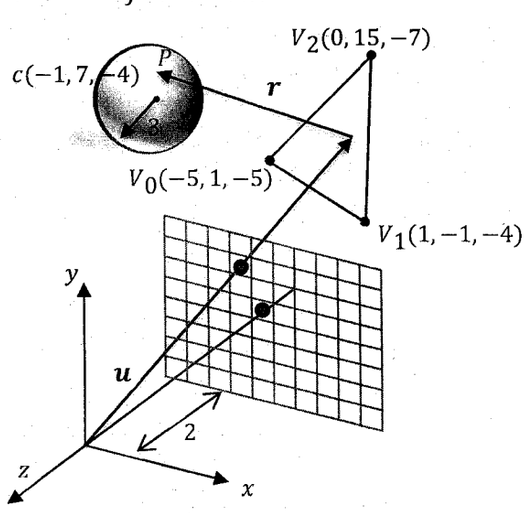


3. a) There are two monochromatic light sources L^0 and L^1 in a scene. The light intensities and corresponding material's reflectivity at vertex V are given in the following table. (12)

	Ambient	Diffuse	Specular
L^0	0.1	0.9	0.9
L^1	0.2	0.7	0.5
V	0.2	0.6	1.0

Furthermore, the shininess at $V = 3.0$, vertex normal at V is $[0 \quad 1 \quad 0]^T$, emissive light at V is 0.05. The position vectors of L^0, L^1, V and eye are, respectively, $[0 \quad 5 \quad 5]^T$, $[5 \quad 5 \quad 5]^T$, $[0 \quad 0 \quad 5]^T$ and $[0 \quad 0 \quad 0]^T$. Use Blinn-Phong illumination model to approximate the total intensity at V.

   b) Consider, the information given in the following figure. Determine whether the reflected ray $\boldsymbol{r}$ intersects the sphere at P. If so, find the P. Here, it is given that the approximated triangular surface normal $N = -1.04i + 1.77j + 9.79k$. (12)



   c) Define and explain the basic radiosity equation. State a method to solve the equation. (11)

4. a) Find Camera Coordinate System (CCS) value for a world point $P(1, 2, 3)$ if the camera is at $(3, 3, 7)$ focusing towards $(1, 1, 1)$ with up vector $U = i + j$. (12)
   b) Write down the 4×4 rotation matrix M that transforms the orthonormal 3D vectors $\boldsymbol{u}(x_u, y_u, z_u)$, $\boldsymbol{v}(x_v, y_v, z_v)$, and $\boldsymbol{w}(x_w, y_w, z_w)$ to orthonormal 3D vectors $\boldsymbol{a}(x_a, y_a, z_a)$, $\boldsymbol{b}(x_b, y_b, z_b)$, and $\boldsymbol{c}(x_c, y_c, z_c)$ such that $M\boldsymbol{u} = \boldsymbol{a}$, $M\boldsymbol{v} = \boldsymbol{b}$, and $M\boldsymbol{w} = \boldsymbol{c}$. (16)
   c) Define z-fighting problem. Mention some strategies to overcome the issue. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define horizontal and vertical retrace. (04)
   b) If a circle's origin is centered at point (5, 5) and radius is equivalent to 5, then rasterize the circle using midpoint circle drawing algorithm. (14)
   c) If $(x_0, y_0) = (5, 15)$ and $(x_{end}, y_{end}) = (11, 9)$, use Bresenham's algorithm to compute pixels for line segment. (09)
   d) What are the major side effects of scan conversion? Explain. (08)

6. a) Write down some limitations of z-Buffer algorithm. Explain how A-Buffer method overcomes the drawback of z-Buffer. (08)
   b) Explain the basic idea of Painter's algorithm. How does the algorithm deal with the situation when the polygons are such that their x, y and z-extents overlap? (10)
   c) Consider two triangles, ΔABC defined by the points $A(1, 7, 5)$, $B(1, 1, 5)$, $C(6, 1, 5)$ and ΔDEF defined by the points $D(6, 5, 2)$, $E(2, 0, 3)$, $F(7, 0, 7)$. Apply z-Buffer algorithm to the triangle surface using a 8×8 depth buffer. Show z-Buffer contents for all the pixels. Also show the frame buffer content if the surface colors of ΔABC and ΔDEF are RED and GREEN respectively. (17)

7. a) Equation of two curves Q and R are given as follows: (09)

$$Q(t) = (1 + t^2)P_1 + (5t - 5t^2)P_2 + t^2P_3$$
$$R(t) = (1 + t^2)P_3 + (5t - 5t^2)P_4 + t^2P_5$$

   For both the curves $0 \leq t \leq 1$. Are both the curves C^0 continuous at $Q(1)$ and $R(0)$? Are both the curves C^1 and G^1 continuous at $Q(0)$ and $R(0)$? Why or why not? Explain your answer.
   b) Show that a cubic Bézier curve can be presented as following matrix formulation; symbols have their usual meanings. (14)

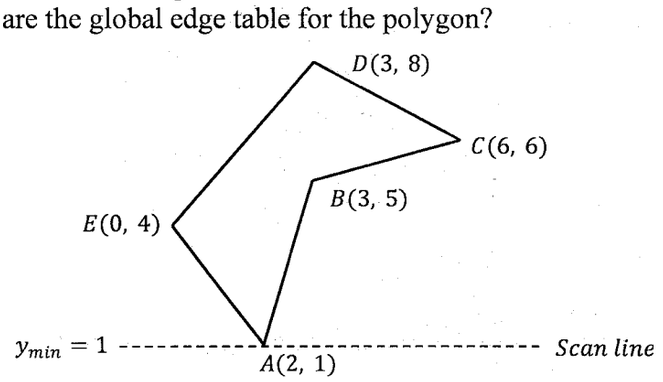
$$P(t) = [t^3 \quad t^2 \quad t \quad 1] \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_1 \\ P_{C_1} \\ P_{C_2} \\ P_2 \end{bmatrix}$$

   Also draw a rough sketch of the curve if $P_1(0, 0)$, $P_{C_1}(0, 3)$, $P_{C_2}(2, 0)$ and $P_2(4, 0)$.
   c) An open uniform B-spline curve $Q(u)$ of order $K = 3$ is constructed with the control points $P_0(0, 0, 0)$, $P_1(1, 1, 0)$, $P_2(2, 0, 0)$, $P_3(3, 1, 0)$ and $P_4(4, 0, 0)$. Determine the knot vector and parametric range for each segment of the curve. Also determine the blending functions for the last segment of the curve. (12)

8. a) How does the texture magnification effect and texture minification effect occur? How to determine the appropriate mipmap level in mipmapping? (05)
   b) How does a CRT monitor display color pictures? Briefly explain. (07)
   c) Define fractal. Generate a fractal image using the following language. (12)

$$AA[A[B]AA(B)]AAAA(A[B]AA(B))$$

   where Alphabet: $\{A, B, [,], (,)\}$,
   Rule: $A \Rightarrow AA$, $B \Rightarrow A[B]AA(B)$, $[] \Rightarrow left\ brance$, $() \Rightarrow right\ brance$.
   d) Consider the scan line Fill algorithm. Determine the content of active edge table to fill the polygon. What are the global edge table for the polygon? (11)



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TIME: 1.5 hours FULL MARKS: 120

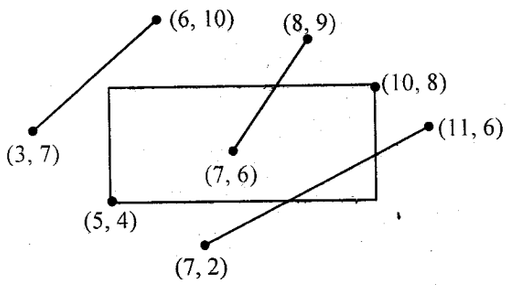
N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Find mirror reflection of a point $P(3, 3, 3)$ about a plane defined by the vertices (1, 0, 0), (0, 1, 0) and (0, 0, 1). (14)
   b) Let a View-Coordinate System (VCS) point $P(1, 1, 7)$. Find the Normalized Device Coordinate System (NDCS) value of P, if the view volume is defined by the following parameters and projection reference point is at VCS origin: (16)
   $left = -1$, $right = 1.2$, $top = 1$, $bottom = -0.8$, $near = 1$, $far = 8$.

2. a) Using Liang-barsky method clip the lines against the clipping rectangle shown in figure below. (16)



   b) Find the parallel projection of a point $P(5, 7, 8)$ with DOP, $\bar{v} = I + 2J + 2K$ onto a plane with plane normal $N = K$, and $P_0(0, 0, 1)$. (14)

3. a) Let a plane defined by the points (0, 0, 0), (0, 0, 2) and (5, 1, 1) is illuminated with a directional white light with direction vector (3, –3, 1). Find the diffuse color for all vertices and specular color for the last vertex by using Phong illumination model. Consider following parameters: (15)
   diffuse reflectivity = (0.4, 0.6, 0.1), *specular reflectivity* = (0.8, 0.9, 0.9), *shininess* = 8, *viewer position* = (6, 2, 3), *vertex normal* = *surface normal*, *no ambient light* and *no attenuation*.
   b) Determine if a ray with $s = I + J + 2K$ and $d = -2K$ intersects the triangle defined by the points (1, 0, 1), (0, 2, 1), (0, 1, 2). Use Badouel's algorithm to verify. (15)

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Define horizontal and vertical retrace. (06)
   b) How does a CRT monitor display color pictures? Briefly explain. (09)
   c) If a circle's origin is centered at point (10, 10) and radius is equivalent to 8, then rasterize the circle using Bresenham's circle drawing algorithm. (15)

5. a) Equation of two bezier curves Q and R are given as follows: (08)

$$Q(t) = (1 + t^2)P_1 + (5t^2 - 5t)P_2 + t^2P_3$$

$$R(t) = t^2P_1 + (5t^2 - 5t)P_2 + (1 + t^2)P_3$$

For both curves $0 \le t \le 1$.

Are both curves C^0 continuous at $Q(0)$ and $R(1)$. Are they both C^1 and G^1 continuous at $Q(0)$ and $R(0)$? Why or why not? Explain your answer.

   b) Find the equation of a Hermite cubic spline with the two end points as (1, 1) & (6, 5) and corresponding tangent vectors as (0, 4) & (4, 0). (13)
   c) How is Koch curves generated? Explain with suitable example(s)? (09)

6. a) How is painting priority of a surface determined by painter's algorithm? (08)
   b) Explain z-buffer algorithm in detail for visible surface detection. What are the advantages and disadvantages of z-buffer algorithm? (13)
   c) Define texture minification. Why is mipmapping used in computer graphics? Briefly explain. (09)

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Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

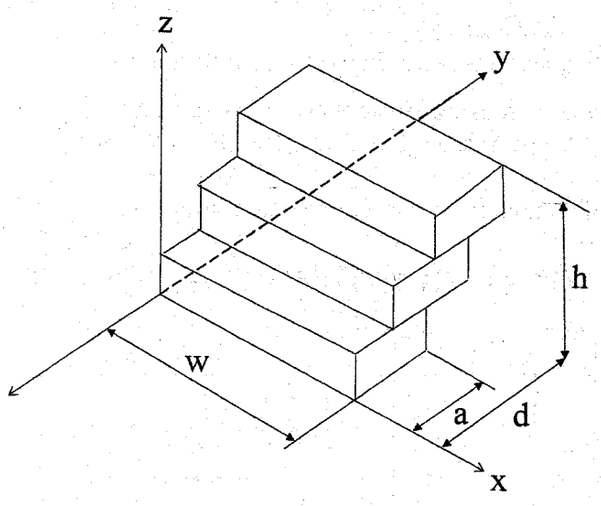
1. a) Define and mention the importance of following transformation in a Computer graphics system. (10)
   i) Modelling transform.
   ii) Viewing transform.

   b) In 2D, it is possible to rotate data using a sequence of a shear in x, followed by a shear in y, and then a non-uniform scaling transform. Rather than using the rotation matrix, you are instructed to use the above mentioned method to perform a θ° rotation. In that case- (15)
   i) What should the X and Y scale factor be?
   ii) Determine the X and Y shear terms and their relationship. As a reminder, the homogeneous shear matrices are given below.

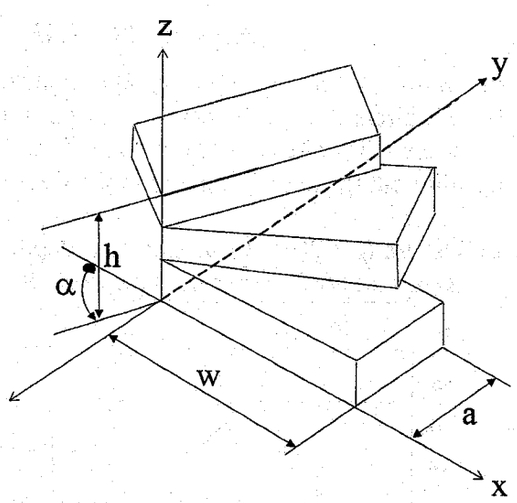
$$shear(x) = \begin{bmatrix} 1 & Sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad shear(y) = \begin{bmatrix} 1 & 0 & 0 \\ Sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

   c) Let a triangle is defined by the points A (1, 0, 0), B (0, 2, 0) and C (0, 1, 1). Determine if a ray with $\mathbf{s} = i + j + 3k$ and $\mathbf{d} = -2k$ intersects the triangle or not? First find the intersection of ray and triangular plane and then use Badouel's algorithm to verify that the intersection lies within the triangle edges. (10)

2. a) Suppose you have to draw a stair having N steps, each step has width w and depth a; the total stair height is h; for a straight forward stair, the total depth is d. See the left figure below, where a stair with N = 3 is shown. Assume that a function *Drawstep(M)* is available to draw a step, subject to a homogeneous transformation M. A step in local coordinates is a cube $-1 \le x,y,z \le 1$, centered around the origin. Furthermore, suppose that a standard set of basic modeling transformation functions for translation, rotation and scaling are available. (11)
   i) Define a matrix M_i using basic transformation functions along with their parameters to draw step i, i =1, …, N for the straight forward stair.
   ii) Consider a spiral stair (see the right Figure below), where the total rotation is α. Define the matrix M_i again for drawing a step i, i =1, …, N for this spiral stair.



Straight Forward Stair



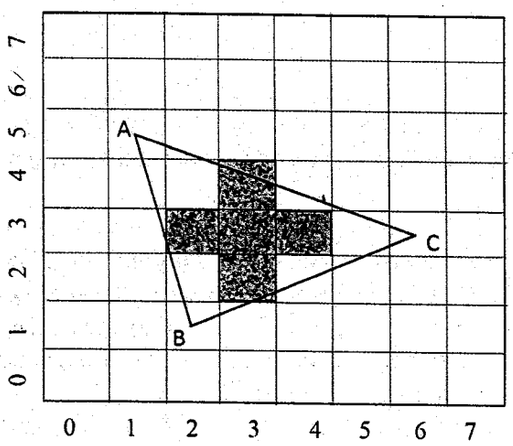
Spiral Stair

   b) Find mirror reflection of a point P (2, 2, 1) about a plane defined by the points (1, 0, 1), (1, 0, 0), and (0, 1, 0). (12)

c) Derive a transformation matrix that transform a world coordinate point to camera coordinate. Be sure to explain how the rotation matrix involved in above transformation is derived. Also consider the camera is defined by the three parameters *position, look-at* and *up-vector.*

3. a) Compare between perspective and axonometric projection. (05)
   b) Derive a projective transformation matrix that perspectively project any point in view-coordinate system (VCS) onto a projection plane parallel to xy-plane and placed at distance –d. Consider the center-of-projection (COP) is at C (x_c, y_c, z_c). (12)
   c) A triangular plane defined by the points (0, 2, 0), (3, 3, 0) and (1, 0, 1) is illuminated by a monochromatic point light source with light intensity 0.9 and positioned at (4, 6, 2). Find the vertex intensities using Phong illumination model if the viewer is at (1, 1, 5), diffuse and specular reflectivity co-efficient of the surface are 0.6 and 0.8 respectively. Also, consider the shininess factor = 8, no ambient light, radial attenuation constraints $a_0 = 0$, $a_1 = 1$, $a_2 = 0$, and vertex normal equals to the surface normal. (15)
   d) State the necessity of near clipping plane while defining a view volume. (03)

4. a) Explain how the Ray tracing method integrates the process of projection, scan-conversion and shadow generation. (10)
   b) Let a triangular face is rasterized to pixel coordinates A (1, 5), B (2,1), C (6,3). Say, RGB vertex intensities are I_A (0.9, 0.1, 0.2), I_B (0.1, 0.8, 0.2) and I_c (0.2, 0.3, 0.7). Use Gauraud shading method to render the triangular face showing the interval pixel intensities of the highlighted pixels shown in figure below. (14)



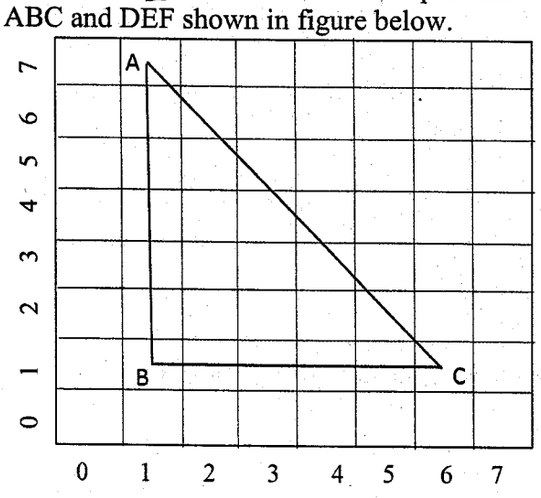
   c) Formulate a transformation matrix that converts an asymmetric parallelepiped view volume to a canonical view volume. Also mention, why should we do the conversion? (11)

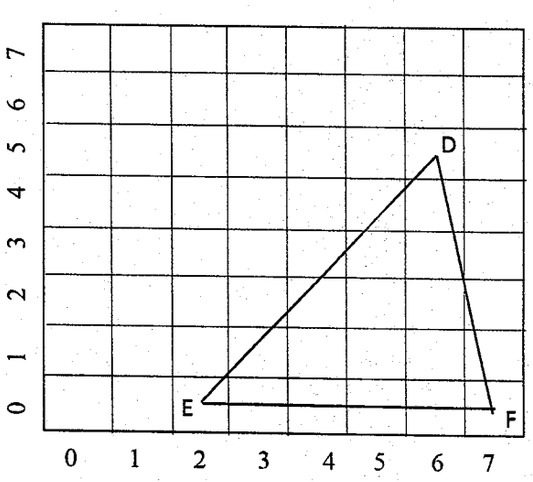
SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Digitize a line (3, 12) to (6, 2) on a raster screen using Bresenham's straight line algorithm. (10)
   b) If a circle's origin is centered at point (30, 40) and its radius is equivalent to 8, then rasterize the circle with midpoint circle algorithm. (10)
   c) Find out the decision parameter values of a quadrant of an ellipse using midpoint ellipse algorithm. (15)

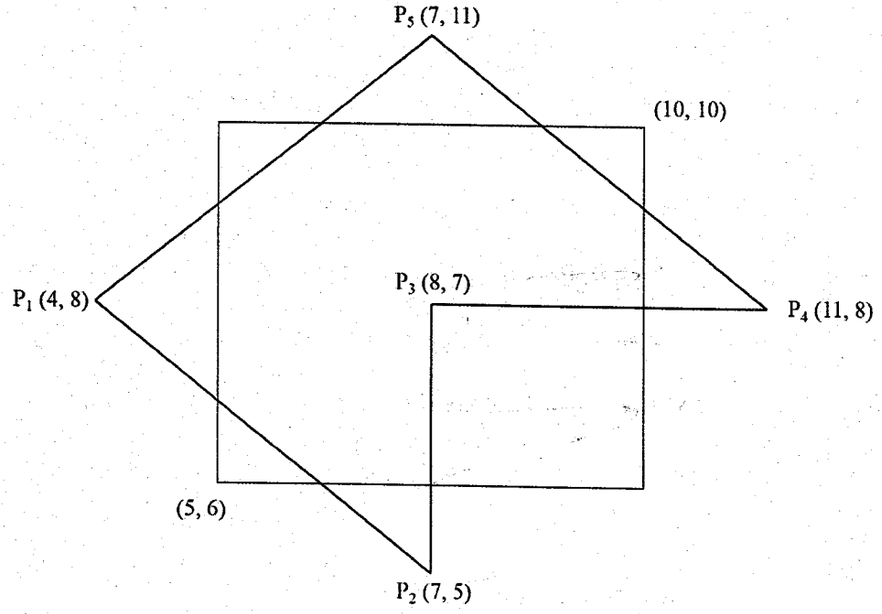
6. a) Use an 8x8 Z-buffer to store the depth values after applying Z-buffer algorithm onto the triangles ABC and DEF shown in figure below. (17)





Here the z-coordinates of the points A, B, C, D, E and F are 5, 5, 5, 2, 3 and 8 respectively. Find the internal Z-values of the triangle before applying the algorithm and show Z-buffer contents after each step. Also show the frame buffer content if the triangle surface of ABC and DEF are RED and GREEN color respectively.

b) Consider the polygon defined by the points P_1, P_2, P_3, P_4 and P_5 as shown in figure below. Use Sutherland-Hodgeman algorithm to clip polygon against the rectangular window defined by the point $(x_{min}, y_{min}) = (5, 6)$ and $(x_{max}, y_{max}) = (10,10)$. Compute the final points list. (13)



c) Define fractal. Draw the first three iterations of Koch snowflake fractal. Show each iteration separately. (05)

7. a) Write about the differences between Object-space method and Image-space method. (08)
   b) How painting priority of a surface is determined by painter's algorithm? (10)
   c) What are the major mapping methods? Give description of them. (07)
   d) Give description of Linear Texture and Cylinder Texture mapping in 2d. (10)

8. a) What is aliasing problem? How one can solve it? (10)
   b) Give an example of special case in scan-line polygon filling algorithm. (05)
   c) What do you know about geometric continuity and parametric continuity? Explain C^0, C^1, C^2 (10)
   d) Calculate Hermite Blending Functions from cubic Hermite Spline Curve having four control points. (10)

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B.Sc. Engineering 4th Year 2nd Term Examination, 2018
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is computer graphics deal with? Mention some prominent applications of it. (06)
   b) Explain the geometry stage of graphics rendering pipeline with necessary illustration. (12)
   c) Find a transformation matrix that aligns line L_1, with line L_2 given by the following equations: (17)

$$L_1 = \begin{cases} \bar{v} = 2I - 3J + 4K \\ P_0 = (1,1,0) \end{cases}$$

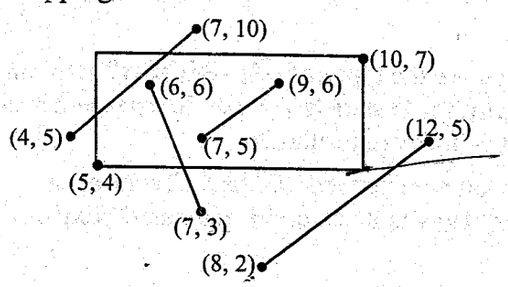
$$L_2 = \begin{cases} x = 1 + 3t \\ y = 4t \quad 0 \le t \le 1 \\ z = 2 - 5t \end{cases}$$

2. a) Derive a transformation matrix to convert an asymmetric view frustum to a Canonical View Volume (CVV). (15)
   b) What is projection? Find a parallel projection of the point (7, 8, 9) with DOP, $\bar{v} = I + 2J + K$ onto a projection plane parallel to xy plane and cuts z-axis at a distance of 1. (10)
   c) What are the parameter needed to define a View-Coordinate System (VCS)? How can a world coordinate point be converted to VCS point? (10)

3. a) Let a VCS point $p(3, 2, 6)$, projection reference point is at VCS origin, and a symmetric view volume is defined by the following parameters: (14)

$$left = right = 1,\ bottom = top = 0.8,\ near = 1,\ far = 7$$

   Find the Normalized Device Coordinate System (NDCS) value of the point p.
   b) Clip the lines presented in the following figure using Cohen-Sutherland algorithm. Also mention how CVV helps 3D clipping. (12)



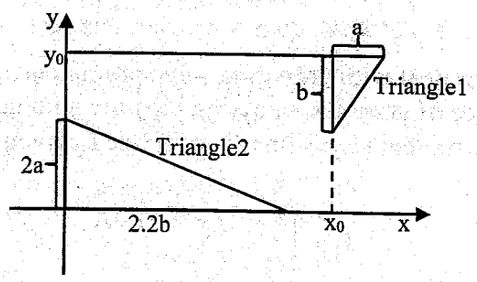
   c) Show that the path of a ray transmitted through a surface is $t = z - \left(\sqrt{1 - |z|^2}\right).n$, where (09)
   $z = \frac{\eta_1}{\eta_2}(u - (u.n))n$, u is an incoming ray, n is the surface normal, and η_1, η_2 are the refractive indices of the media.

4. a) "Phong shading is more realistic than Gouraud shading, but expensive"– Justify the statement. (12)
   b) Let a surface is illuminated by an RGB light source with light intensity (0.9, 1, 0.9), positioned at (5, 4, 2). A vertex (4, 1, 3), whose surface normal is (0, 2, –1), is viewed from (3, 2, 2). Find the intensity that the viewer will see at the vertex, if the surface diffuse and specular reflectively are (0.4, 0.3, 0.6) and (0.8, 0.8, 0.8) with shininess = 10. Consider there is no ambient light, and no attenuation of light. (15)
   c) Explain a method to find the intersection of an incoming ray with a sphere. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07)
   b) Convert these homogeneous points to Cartesian: (0, 1, 3, 2) and (2, 1, 3, 4). (07)
   c) Transform Triangle1 to Triange2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D Homogeneous Coordinates. Compute the final transformation matrix. Note that the size of the triangle has also been changed; see the following figure for the dimensions. (14)



   d) Explain the difference between "Object Coordinates", "World Coordinates", and "Camera Coordinates". (07)

6. a) What are the three major efficiency ingredients in Bresenham's line drawing algorithm? (07)
   b) Consider the line from (0, 0) to (–8, –4), use general Bresenham's line algorithm to rasterize this line. Evaluate and tabulate all the steps involved. (12)
   c) Write down the difference between DDA line drawing algorithm and Bresenham's line drawing algorithm. (06)
   d) Explain the steps in Bresenham's circle drawing algorithm. (10)

7. a) How many real values are necessary to describe the shape of a cubic rational Beizer curve in three dimensional space? Provide a short justification for you answer. (06)
   b) Prove that $R(\theta_1).R(\theta_2) = R(\theta_1 + \theta_2)$, where $R()$ is a rotation function and θ_1, θ_2 are angle of rotation. (08)
   c) Define Hermite Surface. Find the geometry vector G_{Hx} for Hermite Surface. Also find the first of the vecrtor. (10)
   d) Construct the Beizer,curve of order 3 with 4 polygon vertices A(0, 1), B(3, 3), C (4, 5) and D (7, 5) (11)

8. a) What is fractals? What are the types of self-similarity? Explain with example(s). (08)
   b) Explain z-Buffer algorithm in details for visible surface detection. What are the advantages and disadvantages of z-Buffer algorithm? (13)
   c) What is continuity? Compare between G^1 and C^1 continuity. (06)
   d) What is Koch Curves? How is Koch curve generated? Explain with suitable example(s)? (08)

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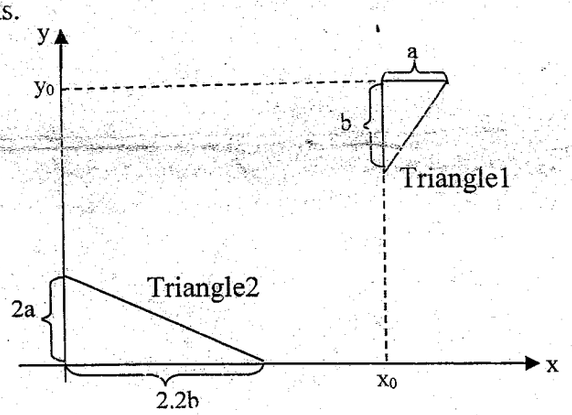
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

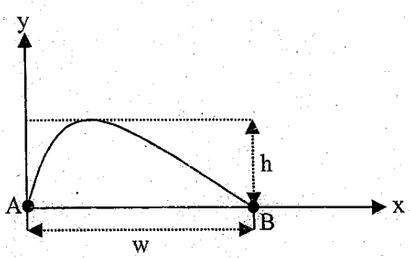
1. a) Explain the differences between "object coordinates", "world coordinates", and "camera coordinates". (08)
   b) When can you convert a quadratic Beizer curve to a cubic Hermite curve? (07)
   c) In Breshenham's algorithm, we saw the efficiency of incremental evaluation. For the function, $f(x) = y = x^2 - 3$, compute the incremental evaluation function. (10)
   d) If $(x_0, y_0) = (5,4)$ and $(x_{end}, y_{end}) = (8,10)$, use Breshenham's algorithm to compute pixels for the line segment. (10)

2. a) Prove that $R(\theta_1).R(\theta_2) = R(\theta_1 + \theta_2)$ where $R()$ is a rotation function and θ_1, θ_2 are angle of rotation. (07)
   b) Transform Triangle1 to Triangle2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D homogeneous coordinates. Compute the final transformation matrix. Note that size of the triangle has also been changed; see the figure for the dimensions. (14)



   c) Reflect the diamond-shaped polygon whose vertices $A(-1,0), B(0,-2), C(1,0)$ and $D(0,2)$ about the line $y = x + 2$. (14)

3. a) We want to draw a cubic Beizer curve starting from $A = (0,0)$ to the point $B = (w,0)$. At A the curve is tangent with y-axis and at B the curve is tangent with x-axis. (see figure below). Draw control points and give the possible positions of them (ignoring h), use new parameters where necessary. (10)



   b) Define fractal. What are the classifications of fractal objects? How is fractal generated? (09)
   c) Find fractal dimension of Koch curve. (05)
   d) Explain depth-buffer algorithm in details for visible surface detection. What are the advantages and disadvantages of z-buffer algorithm? (11)

4. a) How is painting priority of a surface determined by painter's algorithm? (09)

b) A Beizer curve is to be drawn given the control points $P_1(40,40), P_2(10,40), P_3(60,60), P_4(60,0)$. Calculate the coordinates of the points on the curve corresponding to the parameter $t = 0.2, 0.4, 0.6$. Draw a rough sketch of the curve and show coordinates of the various points on it. (14)

c) What is the advantage of using curved surfaces for rendering? How to display it efficiently in practice? (07)

d) What are the drawbacks of Digital Differential Analyzer (DDA) algorithm? (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the three stages of computer graphics pipe-line? Briefly explain each of them. (10)

b) Let a line L, defined in parametric form below. Determine a rotation matrix about L with rotation angle $= 30°$. (14)

$$L: \begin{cases} x = 2 + 3t \\ y = 4t \\ z = 3 + 4t \end{cases} \qquad 0 \le t \le 1$$

c) When performing Cohen-Sutherland line clipping, how do we use outcodes to check for trivial rejection (what exact operation and comparison would you use?) (06)

d) What is GPU? Can it be used in Phong shading? Why? (05)

6. a) Explain the importance of projection in computer graphics. Find a parallel projection matrix with DOP $V = aI + bJ + cK$ and xy as projection plane. (10)

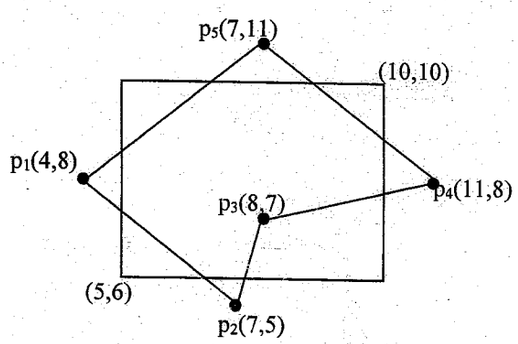
b) Consider a point $P = (2,3,4)$ in eye space. Let a symmetric projection window is defined with aspect ratio $= 1.25$, fovy $= 60°$, near and far plane of the view volume is at a distance 1 and 5 respectively. Find the perspective transformation of the point P in normalized device coordinate space. (12)

c) What is the importance of maintaining pseudo depth in perspective transformation? (05)

d) Describe the anomalies of perspective projection. (08)

7. a) What is view volume? Describe how a parallelepiped view volume can be converted to canonical view volume. (10)

b) Consider the polygon defined by the points p_1, p_2, p_3, p_4, and p_5 as shown in figure below. Use Sutherland-Hodgeman algorithm to clip polygon against the rectangular window defined by the point $(x_{min}, y_{min}) = (5,6)$ and $(x_{max}, y_{max}) = (10,10)$. Compute the final points list. (15)



c) In ray tracing a large computational cost is associated with determining ray-object intersections. Suggest a method that may reduce this cost. (10)

8. a) The normal at a vertex $(0,1,2)$ on a surface is $(0,4,5)$. The light source is at $(0,1,4)$. The diffuse color of the light is $(0.9,0,0.2)$. The diffuse reflectivity of the surface are $(0.4,1,0)$. What will the RGB color of the vertex be? Assume there is no specular, emmisive or ambient component or light attenuation. (10)

b) Explain how light intensity attenuation can be modeled. (07)

c) Let S, be a sphere of radius 10 centered at $(10,-2,-5)$. Determine if a ray $S = 2J + 5K$ and $d = I - 2K$ intersects the sphere. (10)

d) Compare the working principle of Phong and Gouraud shading. (08)

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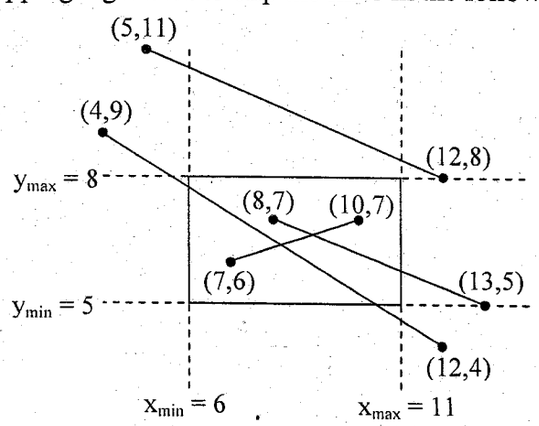
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What does computer graphics deal with? We have cameras to shoot photographs or videos, then what are the driving factors of computer graphics? (07)
   b) Let a unit square defined by $0 \le x \le 1$, and $0 \le y \le 1$ in object coordinate system. New positions X′ are derived from the local positions X by X′ = MX, where M is a transformation matrix. Give a matrix M, expressed in terms of basic modeling transformations, to obtain a rectangle with width = 2 and height = 3, centered around the origin of world coordinate system, and rotated over an angle of 30°. (13)
   c) Why are matrix representations used to describe point transformations in computer graphics? Derive the basic rotation matrix for a rotational transformation about z-axis. (10)
   d) Explain the properties of rigid body transformation and affine transformation. (05)

2. a) Find the mirror reflection of the point (3, 1, 6) with respect to the plane defined by the points (0, 1, 0), (0, 0, 1), and (1, 0, 0). (14)
   b) Explain the working principle of 2D Sutherland-Hodgeman polygon clipping algorithm with necessary illustrations and equations. How can it be extended to 3D? (13)
   c) Briefly explain the taxonomy of planar geometric projection. (08)

3. a) Let the view coordinate system (VCS) is defined by the parameter *eye* $(0, 0, 1)_{WC}$, *look-at* $(0, 0, 0)_{WC}$, *head-up* $\overline{V} = j$. If the projection plane is parallel to the xy plane of the VCS and cuts the negative z-axis of VCS at a distance $d = 1$, find the perspective projection of the point $(1, 1, -1)_{WC}$ on the projection plane considering the VCS origin as a center of projection. (14)
   b) Use Liang-Barsky line clipping algorithm to clip the lines in the following figure. (14)



   c) Briefly explain principal vanishing points for standard perspective projection. Use necessary diagram. (07)

4. a) Let a triangular facet, tessellated part of a large curved surface, is defined by the vertices A (1, 4, 0), B (5, 6, 0), C (4, 0, 0) and their vertex normal are $N_A = -i + j + 3k$, $N_B = j + 3k$, $N_C = i + 4k$ respectively. If the triangle is illuminated by a monochromatic point light source positioned at (–2, 3, 2), and the diffuse reflectivity of the surface is 0.7 then find the diffuse reflection intensity of point P (3, 3, 0) that a viewer will see from position (6, 3, 1). Assume any missing values (if necessary). (15)
   b) Compare among flat shading, Gouraud shading, and Phong shading techniques. (09)
   c) Show that the path of a refracted ray is $\boldsymbol{t} = z - \left(\sqrt{1-|z|^2}\right)\boldsymbol{n}$, where $z = \frac{\eta_1}{\eta_2}(\boldsymbol{u} - (\boldsymbol{u}.\boldsymbol{n}))\boldsymbol{n}$, $\boldsymbol{u}$ is incoming ray, $\boldsymbol{n}$ is surface normal, and η_1, η_2 are the refractive indices of the media. (07)
   d) What are the trade-offs of ray tracing method? (04)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Indicate which raster locations would be chosen by Bresenham's algorithm when scan-converting a line from pixel coordinate (1, 1) to pixel coordinate (8, 5). (12)
   b) What are the major side effects of scan conversion? (08)
   c) Using midpoint algorithm, draw an ellipse with center (2, 2), major axis = 12, and minor axis = 10. (15)

6. a) How do you convert RGB color space into HSI color space? Explain briefly with appropriate figure. (12)
   b) What are the possible representations of a curve? Why is the parametric curve of degree 3 (cubic) preferable? (10)
   c) Define chromatic light and achromatic light. What are the properties of chromatic light? Explain briefly. (07)
   d) What is a raster display? Explain the necessity of scan conversion. (06)

7. a) Define fractal. Generate a fractal image using the following language (12)

   AA[A[B]AA(B)]AAAA(A[B]AA(B))

   where Alphabet {A, B, [,], (,)}

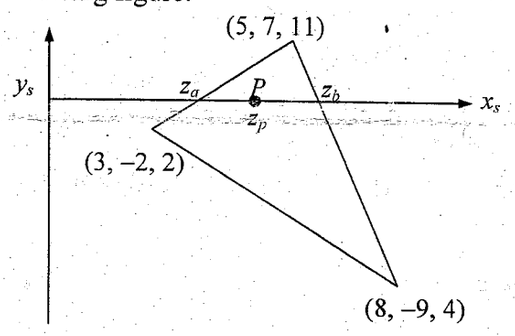
   Rule A ⇒ AA

   B ⇒ A[B]AA(B)

   [] ⇒ left branch

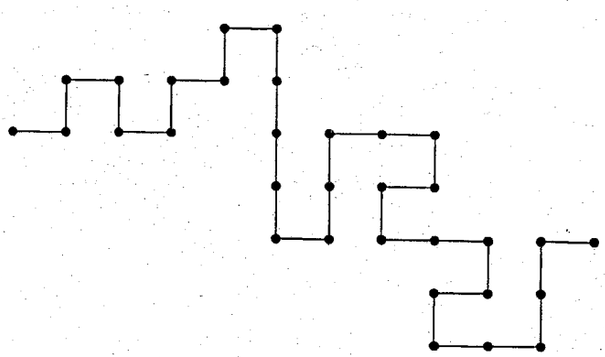
   () ⇒ right branch

   b) What are the difficulties encountered in implementing the painter's algorithm? When does *z*-fighting problem arise? (08)
   c) How does the *z*-buffer algorithm determine which surfaces are hidden and hence compute the z-value at point P of the following figure. (10)



   d) How can you generate the terrain in computer graphics? Explain. (05)

8. a) What do you know about geometric continuity and parametric continuity? Explain G^0, G^1, and G^2 geometric continuity. (10)
   b) Define Hermite surface. Find the geometry vector G_{H_X} for Hermite surface. Also find twist of the vector. (11)
   c) Derive the equation for calculating fractal dimension and hence find the dimension of the following fractals. (07)



   d) What is texturing? Compare between bump mapping and frame mapping techniques. (07)

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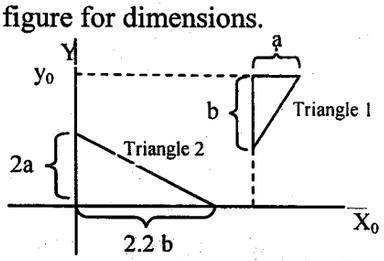
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07)
   b) We wish to scale a triangle A(0,0), B(1,1), C(5,2) to twice its size. While keeping vertex C fixed. This scaling can be expressed as a composite transformation. Describe and write the matrices for the component transformations. (10)
   c) Transform Triangle1 to Triangle2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D Homogeneous coordinates. Compute the final transformation matrix. Note that the size of triangle as also been changed; see the figure for dimensions. (13)

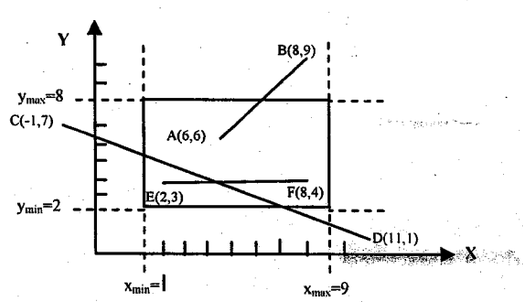


   d) What are the drawbacks of DDA algorithm over Bresenham's line drawing algorithm. (05)

2. a) Scan convert the line segment from point (5,8) to point (9,12). (10)
   b) Why is the point to point line error always $\leq 1/2$ for the mid-point scan conversion algorithm? (10)
   c) Prove that $R(\theta_1).R(\theta_2)=R(\theta_1+\theta_2)$ Where R( ) is a rotation function and θ_1, θ_2 are angle of rotation. (08)
   d) Convert these homogeneous points to Cartesian (0,1,3,2),(2,1,3,4),(2,4,3,5). (07)

3. a) When performing Cohen-Sutherland line clipping, how do we use outcodes to check for trivial rejection (what exact operation and comparison would you use?). (06)
   b) What are the new coordinates of the diamond shaped polygon whose vertices are A(-1,0), B(0,-2), C(1,0) and D(0,2). About the horizontal line y=2 after the reflection. (11)
   c) Derive the perspective projection matrix. In a perspective projection matrices with z_0 =6, scene point (-20,8,15) projects to what (x,y) point on the virtual image plane? To what (x,y) point on the real image plane? (11)
   d) Describe the traditional graphics pipeline and explain what is done in each stage. (07)

4. a) Use the Liang-Bersky algorithm to clip the lines in following figure (12)



b) What are the steps to calculate a transformation matrix to find the mirror reflection of an object with respect to an arbitrary plane whose normal is N and a point P is on the plane. (10)

c) Derive a 3D transformation matrix to rotate an object by θ degree with respect to z-axis. (09)

d) State the name of different types of parallel and perspective projections. (04)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Why do we need illumination models? Derive the Phong specular reflection model and combine diffuse and specular reflection. (13)

b) What is error diffusion? Write the algorithm for error diffusion image representation. (08)

c) Describe the strength and weakness of Ray tracing, using the theory of "The Rendering Equation". (06)

d) Compare between Phong surface rendering method and Gouraud surface rendering method. Which is better and why? (08)

6. a) Consider a triangle which has three vertices as $P_1(1,2)$, $P_2(5,0)$ and $P_3(3,4)$. A scan line S go through it and intersection points are $P_4(3,1)$ and $P_5(4,1)$. Inside the triangle there is a point P(3.5,1) on scan line. Normals for three vertices are $N_1(0,1,0)$,$N_2(0,0,1)$ and $N_3(0,0.7,0.7)$. We also have the following values: L=(0,0.7,0.7), v=(0,0,1) ,K_a=0.1,I_a=0.5,K_d=0.5, K_s=0.2, n_s=1,I_L=0.6, where symbols denote usual meanings. Find Intensity I_4, I_5, and I_p on scan line with Gouraud shading. (22)

b) Prove that reflected ray path, (13)

T=$\frac{n_i}{n_r}u - (\cos\theta_r - \frac{n_i}{n_r}Cos\theta_i)$n, where u is incoming ray and n is normal.

7. a) Let S_1, be a sphere of radius 8 centered at (2,4,1) and S_2 a sphere of radius 10 centered at (10,-2,-5). Determine if a ray with s=2J+5K and d=I-2K intersects the spheres and draw necessary diagram to show the intersections. (14)

b) What do you mean by depth buffer method and z-fighting? (06)

c) Define Hermite Surface. Find geometry vector G_{HX} for Hermite surface. Also find twist of the vector. (10)

d) What is terrain in computer graphics? How do you generate terrain? (05)

8. a) What is particle system? How particles are generated? (07)

b) Define fractal. What are the classifications of fractal objects? How is fractal generated? (09)

c) What do you know about geometric continuity? Explain G^0, G^1 and G^2 geometric continuity? (08)

d) Explain briefly about natural cubic splines and cubic Bezier curves. (11)